

TYUMYATI, Russian Federation

WMO: 218130

Lat: 71.917N

Lon: 123.567E

Elev: 87

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-59.9	-57.3	-63.7	0.1	-59.5	-60.9	0.1	-56.7	19.7	-14.7	18.2	-17.1	0.7	180	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	75.5	61.3	70.5	58.9	66.0	56.6	63.3	73.4	60.1	68.7	57.0	64.3	7.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	74.2	68.6	55.6	66.0	65.0	52.8	59.5	61.3	28.8	73.8	26.6	69.5	24.5	64.4	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	18.9	16.1	DB	-62.0	82.8	4.1	3.3	-64.9	85.2	-67.3	87.2	-69.6	89.1	-72.6	91.5	(4)
(5)			WB	-62.0	65.5	4.1	1.6	-64.9	66.7	-67.3	67.6	-69.6	68.5	-72.6	69.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-32.7	-31.1	-11.7	7.0	23.8	46.5	51.2	49.6	37.0	12.9	-15.2	-28.4		
	DBStd	33.50	14.51	15.83	17.52	12.74	10.79	10.38	8.63	6.36	7.31	11.69	14.65	14.50		
	HDD50	15151	2564	2271	1913	1291	813	190	89	86	397	1149	1956	2432		
	HDD65	20353	3029	2691	2378	1741	1276	563	441	476	841	1614	2406	2897		
	CDD50	289	0	0	0	0	1	84	124	74	6	0	0	0		
	CDD65	19	0	0	0	0	0	7	12	0	0	0	0	0		
	CDH74	192	0	0	0	0	0	71	119	2	0	0	0	0		
	CDH80	35	0	0	0	0	0	11	24	0	0	0	0	0		
	WSAvg	6.5	4.6	5.0	6.6	7.8	8.4	8.2	7.6	7.1	6.4	6.1	5.2	4.9		
	PrecAvg	14.8	1.0	0.7	0.8	0.6	1.1	1.4	1.9	2.0	1.6	1.5	1.0	1.1		
	PrecMax	21.0	2.6	2.4	1.7	1.9	2.3	2.6	4.5	4.0	3.2	2.4	2.2	3.0		
(15) Precipitation	PrecMin	10.6	0.3	0.1	0.2	0.0	0.5	0.3	0.7	0.9	0.7	0.5	0.3	0.3		
	PrecStd	2.5	0.6	0.5	0.4	0.4	0.5	0.6	0.8	0.7	0.7	0.5	0.5	0.7		
	DB	6.5	10.1	29.9	38.4	57.3	82.6	84.1	73.7	67.6	33.9	25.1	14.4			
	MCWB	5.7	9.0	28.7	33.6	47.6	63.3	67.1	61.1	57.0	32.4	24.2	14.3			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	-1.0	2.9	25.1	34.0	48.8	75.4	77.8	70.1	54.8	32.2	18.1	1.8			
	2% MCWB	-1.9	1.9	23.6	30.3	41.6	59.9	64.0	59.0	48.7	31.1	17.1	1.1			
	5% DB	-5.9	-3.0	20.6	30.7	43.0	69.5	71.3	65.8	50.3	29.7	10.7	-2.0			
	5% MCWB	-6.7	-3.5	19.3	27.6	38.0	57.1	60.5	57.2	46.5	28.6	9.7	-2.7			
	10% DB	-10.9	-9.1	15.1	27.0	38.9	64.4	65.8	61.3	46.8	27.4	5.6	-5.7			
	10% MCWB	-11.4	-9.8	13.9	24.5	35.6	54.1	57.1	54.8	43.6	26.2	4.8	-6.3			
	0.4% WB	5.9	9.2	28.4	34.0	48.9	64.8	68.9	62.9	57.2	33.0	24.4	14.2			
	0.4% MCDB	6.4	10.5	29.5	37.5	56.4	79.1	81.9	70.5	67.8	33.5	25.4	14.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-1.7	1.8	23.7	31.1	42.1	61.6	65.1	60.5	50.1	31.1	17.3	1.2			
	2% MCDB	-1.0	2.8	24.8	33.6	48.4	72.5	74.7	68.3	52.8	32.1	18.2	1.5			
	5% WB	-6.6	-3.5	19.3	27.8	38.8	57.9	61.5	58.1	46.6	28.9	9.7	-2.6			
	5% MCDB	-5.9	-3.0	20.6	30.7	42.7	69.1	70.4	64.1	49.5	29.9	10.5	-2.1			
	10% WB	-11.3	-9.7	13.9	24.6	35.7	54.5	57.3	55.2	43.7	26.2	4.7	-6.3			
	10% MCDB	-10.8	-9.2	15.0	27.1	38.7	64.0	63.9	61.0	46.8	27.2	5.5	-5.6			
	MDBR	13.7	13.9	18.6	20.6	12.3	14.5	13.5	13.7	10.1	10.3	13.1	12.9			
	5% DB	16.5	17.3	17.7	20.5	15.8	23.3	21.9	23.5	17.0	8.5	17.1	17.1			
(36) Mean Daily Temperature Range	MCDBR	16.2	16.9	17.4	18.2	9.7	11.9	11.1	13.7	11.3	8.4	16.8	16.8			
	MCWBR	16.8	17.7	18.3	20.2	14.0	22.8	19.9	20.8	15.6	8.9	17.5	17.1			
	MCDBR	16.5	17.3	18.2	18.1	9.0	13.5	11.1	13.6	11.1	8.9	17.3	16.9			
	MCWBR															
(40) Clear-Sky Solar Irradiance	taub	N/A	0.197	0.277	0.308	0.339	0.343	0.391	0.375	0.297	0.205	N/A	N/A			
	taud	N/A	1.958	2.044	2.080	2.139	2.400	2.295	2.302	2.401	1.992	N/A	N/A			
	Ebn at Noon	0	140	213	252	260	266	244	230	219	126	0	0			
	Edh at Noon	0	10	24	34	37	30	32	28	18	9	0	0			
(44) All-Sky Solar Radiation	RadAvg	0	69	525	1324	1794	1727	1524	977	448	110	0	0			
	RadStd	0	8	34	56	92	157	96	114	56	16	0	0			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+1.89	N/A	N/A	N/A	N/A	N/A	-743	-709	N/A	N/A

Nomenclature: See separate page